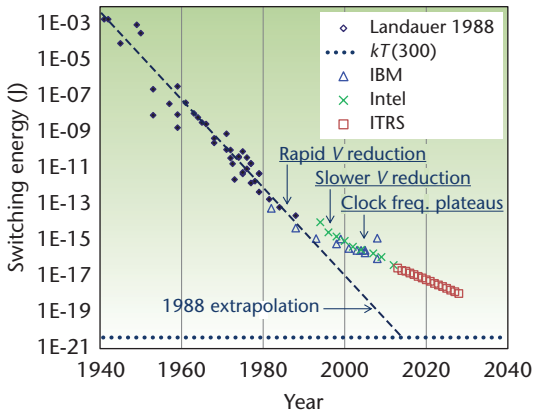


Figure 1. Minimum switching energy dissipation in logic devices used in computing systems as a function of time. Black diamonds replicate data from Rolf Landauer,[5] and the dashed line is Landauer's 1988 extrapolation of the historic trend toward kT (evaluated at $T = 300$ K), indicated by the dotted line. Triangles and Xs are published values from IBM and Intel, respectively, compiled by Chi-Shuen Lee and Jieying Luo at Stanford University during their PhD thesis research with one of the authors (Wong). Open squares are values from the 2013 International Technology Roadmap for Semiconductors (ITRS). The data is available at https://purl.stanford.edu/gc095kp2609. Current, up-to-date data can be accessed at https://nano.stanford.edu/cmos-technology-scaling-trend.

Figure 1 shows a very important exponential trend in information technology that already shows a sharp slowing of progress. In 1988, Rolf Landauer[5] published some remarkable data on energy dissipation in computing that had been collected over many years by his IBM colleague Robert Keyes. From the 1940s through the 1980s, a span that includes the replacement of vacuum tubes by bipolar transistors, the invention of the integrated circuit, and the early stages of the replacement of bipolar transistors by field-effect transistors (FETs), the typical energy dissipated in a digital switching event dropped exponentially by over 10 orders of magnitude. We replotted that data in Figure 1 along with Landauer's extrapolation of the trend toward switching energies on the order of the thermal fluctuation energy, kT, evaluated at $T = 300$ K. Landauer was well aware that the switching energy would not approach kT around 2015, not with the established complementary metal-oxide-semiconductor (CMOS) device and circuit technology. His extrapolation was a way of highlighting the possibility of, and perhaps the need for, a new way of computing.

Some have mistaken the kT per switching event as a fundamental lower bound on the energy consumption of digital computation. Landauer knew that it isn't. His 1988 publication reviewed fundamental research demonstrating the possibility of an energy-conserving form of computation. As in today's circuits, the devices in energy-conserving circuits would store enough energy—many times kT—to reliably distinguish the digital state from the inevitable thermal noise. For good engineering reasons, today's circuits dissipate that stored energy every time a device is switched. In contrast, energy-conserving circuits would dissipate only a small fraction of the stored energy in each switching event. In such circuits, there's no fundamental lower bound on the energy efficiency of digital computation.

Although no commercially viable energy-conserving computing systems emerged in the 1990s or later, digital quantum computing, still in its infancy, exemplifies the energy-conserving approach. To show what did happen in the commercial sector after 1988, we added data to Figure 1 showing switching energies for minimum channel width CMOS FET technologies based on technical publications from IBM and Intel. For a while, switching energy continued to drop rapidly, as IBM led the industry in rapidly reducing operating voltage. Roughly following the elegant scaling rules laid out by Robert Dennard and colleagues,[6] each successive generation of smaller, lower voltage, lower power devices was also faster. Increasingly potent CMOS technology extended the long run of exponential increases in microprocessor clock frequency[7]—a key measure of computing performance—that had begun with the Intel 4004 in 1972. And the ever smaller, ever cheaper transistors enabled rapid elaboration of computer architecture. For example, the introduction in the 1990s of sophisticated approaches to instruction-level parallelism (superscalar architectures) further multiplied the system-level performance gains from increasing clock speed. By the late 1990s, CMOS had displaced the more power-hungry bipolar transistor from its last remaining applications in high-performance computing. Despite these triumphs, the historic rate of reduction in switching energy couldn't be maintained through the 1990s as the FET approached some fundamental constraints to its further development.

Physical Constraints on Continued Miniaturization

Note that this slowing of a highly desirable exponential trend in information technology—the break in slope so evident in Figure 1—has nothing to do with the approach of switching energy toward the thermal fluctuation energy. Even in